Surface Energies of Ternary Ni-Co-Cr Alloys

Group 9 (Ni-Co-Cr)
Shriram Atul Naik (111) Kethavath Mounika (100) Lakshya Rishi (110)
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Abstract

The (111), (100) and (110) surface energies of three random fcc Ni-Co-Cr alloys ($Ni_{22}Co_{39}Cr_{39}$, $Ni_{33}Co_{33}Cr_{33}$, $Ni_{44}Co_{28}Cr_{28}$) were calculated at 300, 500 and 700 K with LAMMPS and the Sharifi-Wick MEAM potential, using 81 production calculations (3 compositions × 3 surfaces × 3 temperatures × 3 random configurations). The surface energy follows $\gamma_{111} < \gamma_{100} < \gamma_{110}$ for every composition and temperature, and falls by 3.6-5.5 % when Co and Cr are replaced by Ni. No temperature dependence of the excess potential energy can be resolved between 300 and 700 K (all mean changes are almost 8 mJ m^{-2}). The absolute (111) values are about two thirds of those of Shuang et al., but the composition trend is reproduced, and pure Ni(111) is about 24 % below the DFT value. The trends, rather than the absolute numbers, are therefore regarded as the credible result.

1. Introduction

CoCrNi is a medium-entropy alloy, valued for its combination of high strength and ductility with good fracture toughness [4]. The surface energy is a basic quantity in the description of fracture, crystal growth and the behaviour of free surfaces and interfaces. In a chemically disordered alloy every surface cut is a different atomic arrangement, so a single cut gives only one sample from a distribution of surface energies: several arrangements have to be sampled and their spread reported. The objective here is to determine how composition, surface orientation and temperature affect the surface energy γ, and to show that the calculated values are physically sensible. The three compositions are those of Shuang et al. [2], whose (111) surface energies serve as the main literature reference. Co and Cr are kept equal so that only the Ni content varies. Random configurations represent the limit without short-range order (SRO).

2. Methodology

2.1. Interatomic potential

The MEAM potential of Sharifi and Wick [3] for Cr-Ni-Co was used (NIST entry `2025-Sharifi-H-Cr-Ni-Co-LAMMPS-ipr1`, cutoff 5.0 Å). Elastic constants of pure Ni were obtained at 0 K from the stress response to small strains (±0.2 %) applied to a relaxed cubic cell, with the atoms re-minimised after each deformation. The lattice parameter and bulk modulus agree with experiment, while the individual elastic constants deviate by 13–17 % (Table 1). Surface energies and thermal properties were not fitting targets of the potential.

Table 1: MEAM lattice parameter and elastic constants of pure Ni compared with experiment.

Property	MEAM	Exp.
a (Å)	3.521	3.52
C_{11} (GPa)	217	261
C_{12} (GPa)	171	151
C_{44} (GPa)	111	132
B (GPa)	186	188

2.2. Surface energy and bulk reference

The surface energy is defined as

$$\gamma = \frac{E_{\text{slab}} - E_{\text{bulk}}}{2A}, \qquad A = L_x L_y, \tag{1}$$

where the factor 2 accounts for the two free surfaces and 1 eV/Å^2 = 16021.8 mJ m^{-2}. In a random alloy the bulk energy depends on the particular arrangement, so E_{bulk} is taken as the energy of the periodic bulk block containing exactly the atoms of the slab, and E_{slab} as the energy of the same atoms after vacuum is inserted. The arrangement-dependent part cancels out and an inconsistent bulk reference, whose error grows with slab thickness is avoided. The quantity is an excess potential energy, not the surface free energy.

2.3. Supercells and configurations

All FCC cells are orthogonal, with z along the surface normal and periodic in x and y. The x, y, z axes are along $[1\bar{1}0]$, $[11\bar{2}]$, $[111]$ for (111); [100], [010], [001] for (100) and [001], $[1\bar{1}0]$, [110] for (110). The box edges must be whole multiples of the repeat length of each orientation, so atom number, area and thickness cannot be exactly identical for the three planes (Table 2). The physical dimensions (thickness ≈28 Å, in-plane edge ≈20 Å, vacuum 20 Å) were therefore matched approximately instead of the layer counts. Each configuration has exact stoichiometry (to within one atom) and a random placement of atoms from a seeded shuffle (seeds 1, 2, 3). The same arrangement is used at all three temperatures.

Table 2: Supercells before vacuum insertion. Areas are for the starting $a = 3.50$ Å; thickness is the distance between the outer layers (height of supercell), (layers − 1) times the interplanar spacing.

	(111)	(100)	(110)
Repeats $n_x \times n_y \times n_z$	$8 \times 5 \times 5$	$6 \times 6 \times 8$	$6 \times 8 \times 12$
Atoms	1200	1152	1152
Layers	15	16	24
Area (Å^2)	424	441	416
Thickness (Å)	28.3	26.3	28.5

2.4. Simulation sequence

Each calculation starts with a 0 K relaxation of atoms and box, followed by an NPT bulk run at the target temperature (10 ps equilibration and 40 ps production, 2 fs time step), from which the mean box is taken for subsequent NVT runs. E_{bulk} is the time-averaged potential energy of an NVT run in this box (10 ps + 40 ps). Vacuum is then inserted along z into the same atoms, and an NVT slab run (10 ps + 40 ps) gives E_{slab}. The statistical error of each run is estimated from 10 block averages and is within 2-8 mJ m^{-2}. The uncertainty reported for each case is the standard deviation (SD) over the three configurations.

3. Convergence

Convergence was examined for $Ni_{33}Co_{33}Cr_{33}$, (111), 700 K (Fig. 1), with slab thickness and in-plane area varied independently. The box lengths and the potential energies of bulk and slab settle within the equilibration window, and the running surface energy stays within about 1 % of its final value during the production run, so the averaging window is sufficient (a-d). In the thickness test (e), the 9-layer slab is clearly too low (by about 4 %), while the slabs of 12-24 layers lie within about 2.5 % of the production value without a monotonic trend. This spread is of the order of the configuration scatter (SD about 1-1.5 %), because every size contains a different random arrangement. A ±3 % band is therefore used in (e) instead of ±1 %, and thickness effects below this level cannot be separated from arrangement noise in this test. The cleaner evidence comes from 0 K calculations for pure Ni, which carry no arrangement noise: they gave identical (111) surface energies for 9-24 layers and for vacuum widths of 10-40 Å. For the in-plane size (f), only the smallest cell, whose edges are shorter than 20 Å, is noticeably low (about 3.5 %), whereas the larger cells agree within about 1 %. The production cell (15 layers, 4.4 nm^2 for the relaxed cell, 20 Å vacuum) was therefore adopted. The tests were performed on (111) only; the (100) and (110) cells were built with the same physical thickness, in-plane edge and vacuum.

4. Predictions

The predictions below come from discussions we had in the group about coordination, bulk properties and the choice of potential. The composition argument was formulated after the (111) results had been seen from Shuang et al. [2], so for (111) it is a consistency check.

1. *Orientation.* $\gamma_{111} < \gamma_{100} < \gamma_{110}$, following the number of bonds broken (3,4 and 5 for the 3 planes respectively). This was only qualitatively predicted.
2. *Composition.* γ decreases as Ni replaces Co and Cr, considering that the alloy might follow the pure-element rule of mixtures for surface energies (Sec. 5.2).
3. *Temperature.* It was expected that γ would decrease with temperature, because the vibrational entropy contirbution increases at higher temperatures.
4. *Magnitude.* Absolute values might be off in magnitude since the potential was not fitted to surface energies.

Table 3 sets these against the results of the next section.

Table 3: Expectations and outcome.

Expectation	Outcome
Orientation	Supported: The exact trend that was predicted was observed.
Composition	Supported: −3.6 to −5.5 % decrease from $Ni_{22}Co_{39}Cr_{39}$ to $Ni_{44}Co_{28}Cr_{28}$; alloys.
Temperature	Not observed: No resolvable trend, mean changes ≤ 8 mJ m^{-2}i.e. the ideal temperature dependence is not captured by the simulations.
Magnitude	Supported: 24 % below DFT, 32-36 % below Shuang et al. but the trends are observed

5. Results & Discussion

5.1. Surface orientation and chemical disorder

The surface energies are given in Table 4 and Fig. 2. For every composition and temperature, $\gamma_{111} < \gamma_{100} < \gamma_{110}$. This follows the coordination of the surface atoms: 9 in the close-packed (111) plane, 8 in (100) and 7 in (110). The fewer the neighbours that are lost at the surface, the lower the energy. The ratios to (111) are almost independent of temperature and composition (Table 5). A simple count of broken nearest-neighbour bonds gives the same ordering: a surface atom loses 3 of its 12 bonds on (111), 4 on (100) and 5 on (110). The simulated γ_{100} and γ_{110} lie 15-17 % and 18-19 % above γ_{111} (Table 5), so the bond count explains the order but not the size of the differences. Interactions beyond the first shell, which a nearest-neighbour count ignores, are a plausible reason, but this was not tested. The (100) and (110) values are close, and for $Ni_{22}Co_{39}Cr_{39}$ they differ by less than the scatter.

Chemical disorder appears as configuration-to-configuration scatter. On (111) it is largest for $Ni_{22}Co_{39}Cr_{39}$ and smallest for $Ni_{44}Co_{28}Cr_{28}$, and it decreases with Ni content, less clearly, on (100); on (110) no clear composition dependence is seen. Three configurations are enough to resolve the trends discussed here.

Table 4: Surface energy (mJ m^{-2}), mean ± SD over three configurations.

Surface	Alloy	300 K	500 K	700 K
(111)	$Ni_{22}Co_{39}Cr_{39}$	1731 ± 34	1734 ± 36	1736 ± 30
	$Ni_{33}Co_{33}Cr_{33}$	1703 ± 23	1696 ± 15	1698 ± 20
	$Ni_{44}Co_{28}Cr_{28}$	1660 ± 3	1660 ± 4	1658 ± 7
(100)	$Ni_{22}Co_{39}Cr_{39}$	2025 ± 29	2022 ± 28	2028 ± 16
	$Ni_{33}Co_{33}Cr_{33}$	1960 ± 13	1956 ± 14	1961 ± 7
	$Ni_{44}Co_{28}Cr_{28}$	1913 ± 12	1912 ± 13	1918 ± 8
(110)	$Ni_{22}Co_{39}Cr_{39}$	2052 ± 19	2056 ± 27	2054 ± 22
	$Ni_{33}Co_{33}Cr_{33}$	2011 ± 13	2007 ± 15	2016 ± 17
	$Ni_{44}Co_{28}Cr_{28}$	1979 ± 10	1975 ± 14	1971 ± 19

5.2. Alloy chemistry and the rule of mixtures

On all three surfaces, γ decreases as Co and Cr are replaced by Ni, from $Ni_{22}Co_{39}Cr_{39}$ to $Ni_{44}Co_{28}Cr_{28}$ by

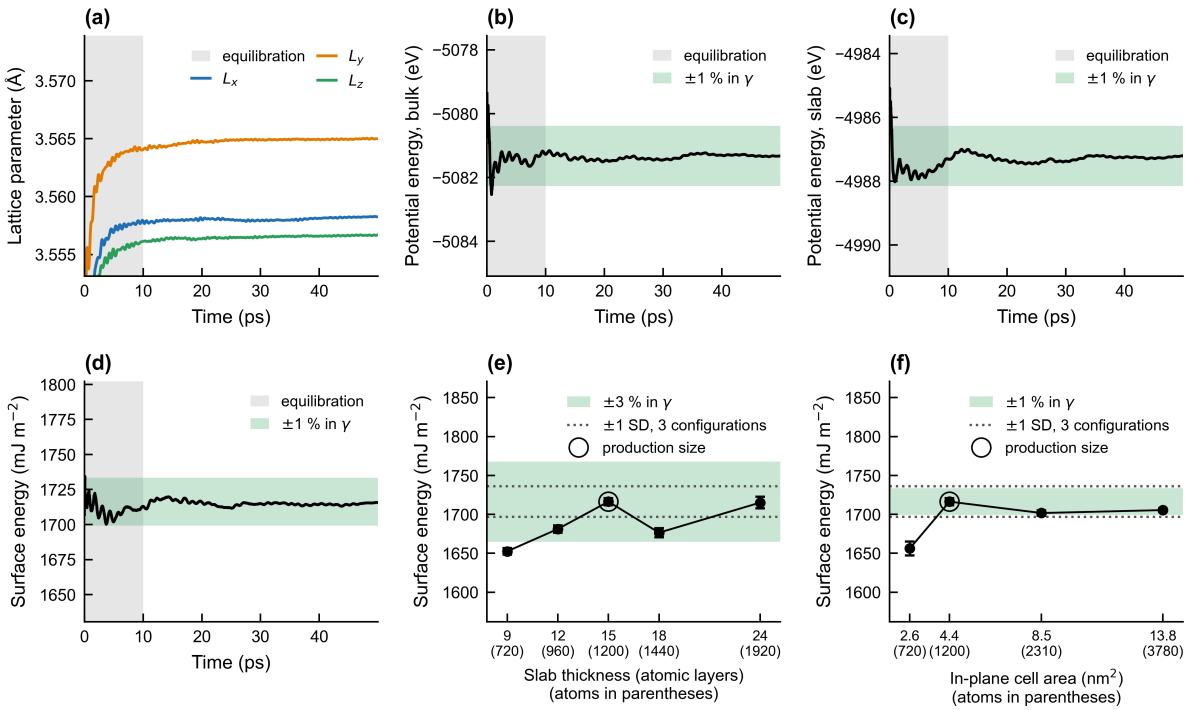


Figure 1: Convergence for $Ni_{33}Co_{33}Cr_{33}$, (111), 700 K. (a) Box lengths divided by the number of repeats (effective lattice parameter) during the NPT run; (b, c) running potential energy of the bulk and slab NVT runs; (d) running surface energy; (e) γ against slab thickness; (f) γ against in-plane cell area. Panels (a-d) use one configuration; each size in (e, f) is a single run with its own random arrangement. The band in (e) is ±3 % because the differences between sizes are limited by configuration scatter; (d) and (f) use ±1 %.

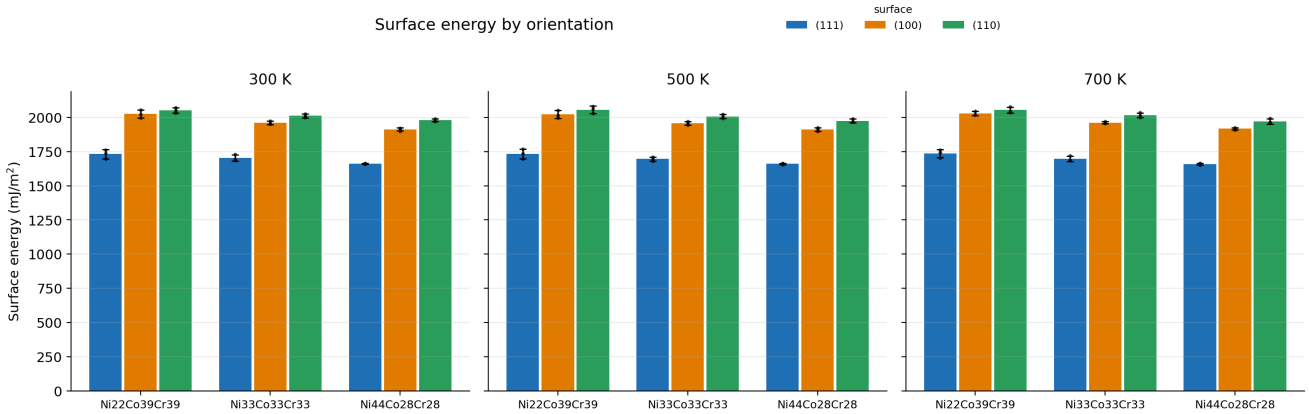


Figure 2: Surface energy by orientation and composition at 300, 500 and 700 K. Bars: mean over three configurations; error bars: ±1 SD; points: individual configurations.

Table 5: Surface energy relative to (111), averaged over 300, 500 and 700 K. The last row is the ratio of broken nearest-neighbour bonds per surface atom.

Alloy	$\gamma_{100}/\gamma_{111}$	$\gamma_{110}/\gamma_{111}$
$Ni_{22}Co_{39}Cr_{39}$	1.168	1.185
$Ni_{33}Co_{33}Cr_{33}$	1.153	1.184
$Ni_{44}Co_{28}Cr_{28}$	1.154	1.190
Broken bonds	$4/3 \approx 1.33$	$5/3 \approx 1.67$

4.1 % on (111), 5.5 % on (100) and 3.6 % on (110) at 300 K. Table 6 gives the full $Ni_{22}Co_{39}Cr_{39}$-$Ni_{44}Co_{28}Cr_{28}$ change with its standard error. On (100) and (110) it is about 6 standard errors and both composition steps have the same sign. On (111) it is only 3.6 standard errors, which is suggestive but not conclusive for three configurations, and the $Ni_{22}Co_{39}Cr_{39}$-$Ni_{33}Co_{33}Cr_{33}$ step (29 mJ m^{-2}) is within one SD. To see whether the trend is consistent with simple additivity, pure fcc Ni, Co and Cr were calculated on (111) at 500 K with the same protocol (single runs, since a perfect crystal has no configurational scatter), giving 1441, 1963 and 1319 mJ m^{-2}. With $x_{Co} = x_{Cr}$, the linear mixture is

$$\gamma = x_{Ni}\,\gamma_{Ni} + (1 - x_{Ni})\,\frac{\gamma_{Co} + \gamma_{Cr}}{2}. \qquad (2)$$

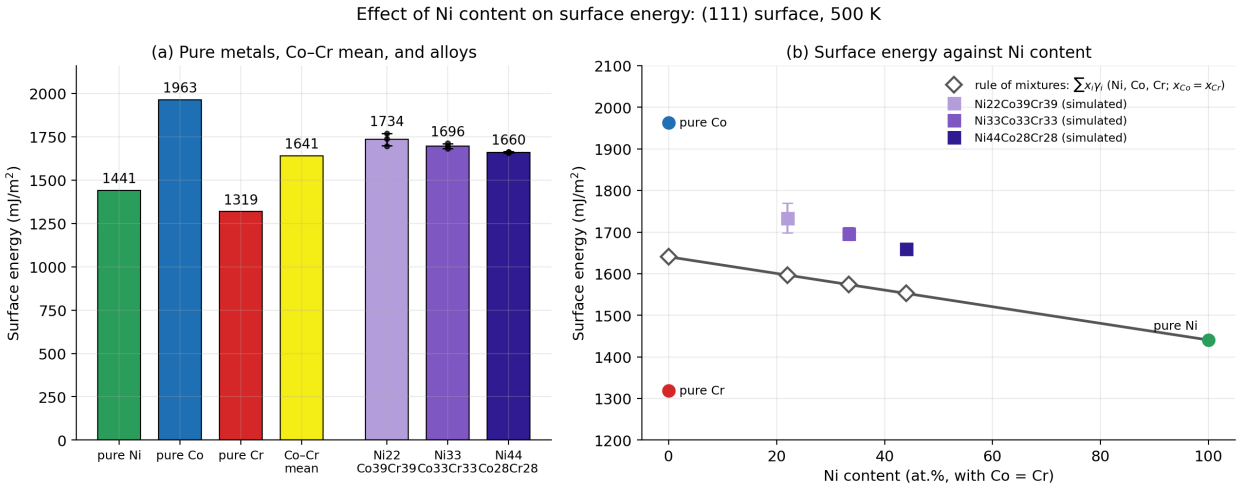


Figure 3: (111) surface at 500 K. (a) Pure metals (single runs), the Co-Cr average of the two pure values (calculated, not simulated) and the three alloys (mean ± SD; points are the configurations). (b) γ against Ni content with Co = Cr and the linear rule of mixtures.

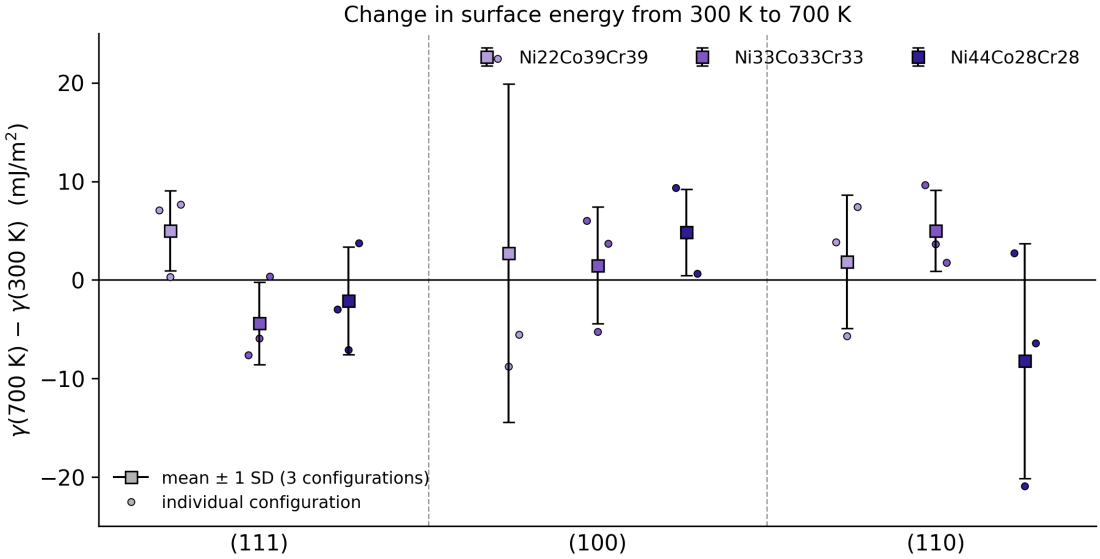


Figure 4: Change of γ between 300 and 700 K for the same atomic arrangement. Squares: mean ±1 SD over three configurations; circles: individual configurations.

Since pure Ni lies below the Co-Cr average (1641 mJ m^{-2}), each Ni atom added lowers the surface energy on this line, which is consistent with the direction of the observed trend (Fig. 3). The rule of mixtures is only a simple additivity check, not a mechanism, and the alloys deviate from it by 7-9 %. The alloys lie roughly 7-9 % above the line (Table 7), so the mixing is not strictly additive. Fcc Co and Cr are hypothetical phases at these temperatures (Co is hcp and Cr bcc), so the pure-element values are model quantities.

Table 6: Change of γ from $Ni_{22}Co_{39}Cr_{39}$ to $Ni_{44}Co_{28}Cr_{28}$ at 300 K (difference of means ± standard error, $\sqrt{(s_1^2 + s_2^2)/3}$). With three configurations per alloy, roughly 3-4 standard errors are needed for 95 % confidence.

Surface	$\Delta\gamma$ (mJ m^{-2})	Std. error	$\lvert\Delta\gamma\rvert$/s.e.
(111)	−71	20	3.6
(100)	−112	18	6.1
(110)	−73	13	5.8

Table 7: (111) at 500 K: rule of mixtures and simulated alloys (mJ m^{-2}).

Alloy	Mixture	Simulated	Difference
$Ni_{22}Co_{39}Cr_{39}$	≈1597	1734	+137 (+8.6 %)
$Ni_{33}Co_{33}Cr_{33}$	≈1575	1696	+121 (+7.7 %)
$Ni_{44}Co_{28}Cr_{28}$	≈1553	1660	+107 (+6.9 %)

5.3. Temperature dependence

Between 300 and 700 K no systematic change of γ is observed (Table 4, Fig. 4). The mean changes are at most about 8 mJ m^{-2}, with either sign. Because the same arrangement is used at every temperature, Fig. 4 shows paired differences; none of the nine is significant ($|t| < 4.3$, the 95 % critical value for two degrees of freedom). This is not surprising, because γ here is an excess energy. The surface free energy is

$$\gamma_F = \gamma_E - \frac{T\,\Delta S}{2A}, \qquad (3)$$

where ΔS is the excess (mainly vibrational) entropy of the slab relative to the bulk. Only the energy term, γ_E, is obtained from Eq. (1). The entropy term is not calculated, so the flat trend does not say much about how the surface free energy changes with temperature.

5.4. Comparison with the literature

Table 8 compares the (111) values with Shuang et al. [2] and with the DFT value for Ni(111) of Tran et al. [1]. The present values are systematically lower than those of Shuang et al., by 32-33 % for the alloys and 36 % for pure Ni, and the pure-Ni value is 24 % below DFT. The deviation is almost uniform across the three alloys, so the composition trend is reproduced: from $Ni_{22}Co_{39}Cr_{39}$ to $Ni_{44}Co_{28}Cr_{28}$ the surface energy falls by 4.1 % here and 3.8 % in Shuang et al., and pure Ni lies below all three alloys in both data sets. Since the potential was not fitted to surface energies, the trends, and not the absolute values, are the meaningful comparison. Also the absolute values are specific to the potential which was different for Shuang et al. [2].

Table 8: (111) surface energies compared with the literature ($mJ\,m^{-2}$). Alloys: present values at 300 K; pure Ni: 0 K, relaxed. The last reference value is DFT. Deviation = (present − reference)/reference.

System	Present	Reference	Deviation (%)
$Ni_{22}Co_{39}Cr_{39}$	1731.3	2573 [2]	−32.7
$Ni_{33}Co_{33}Cr_{33}$	1702.5	2508 [2]	−32.1
$Ni_{44}Co_{28}Cr_{28}$	1659.9	2475 [2]	−32.9
Pure Ni	1450.9	2269 [2]	−36.1
Pure Ni	1450.9	1920 [1]	−24.4

5.5. Sampling and uncertainty

The uncertainty quoted for each case is the standard deviation (SD) over the three configurations. It is between 0.2 and 2.1 % of γ, about 1 % on average, and it is larger than the statistical error of a single run (2–8 $mJ\,m^{-2}$) in most cases. The main source of uncertainty is therefore the different atomic arrangements, not the thermal sampling. Three configurations are enough to show the ordering of the surfaces and the composition trend on (100) and (110), where the change from $Ni_{22}Co_{39}Cr_{39}$ to $Ni_{44}Co_{28}Cr_{28}$ is about 6 standard errors (Table 6). They are not enough to resolve differences of 1–2 % or less, such as the $Ni_{22}Co_{39}Cr_{39}$–$Ni_{33}Co_{33}Cr_{33}$ step on (111) or any temperature effect, and the SD itself is only known to about 50 %. Ten or more configurations would be needed for that. The seeds are fixed, so the results can be reproduced, and the same arrangements are used at all temperatures.

6. Limitations

The main limitation is the potential. The absolute surface energies are 24–36 % below DFT and the values of Shuang et al., because the potential was not fitted to surface energies or thermal properties. Further limitations are:

1. Only the excess potential energy is obtained, not the free energy, so the temperature dependence of the surface free energy cannot be assessed.
2. Three configurations per case give only a rough estimate of the scatter.
3. Short-range order (SRO) is not modelled, and no diffusion or segregation is possible in 50 ps.
4. Convergence was tested on one alloy and one surface, with a single arrangement per size.
5. Pure fcc Co and Cr are hypothetical phases, so the pure-element values are model quantities.

7. Conclusions

Within this model, $\gamma_{111} < \gamma_{100} < \gamma_{110}$ for all three random Ni-Co-Cr alloys between 300 and 700 K, consistent with the lower coordination loss on close-packed planes. Replacing Co and Cr by Ni lowers the surface energy by 3.6-5.5 %, in the direction expected from a rule of mixtures of the pure elements, although the alloys lie above the linear mixture. No temperature dependence of the excess potential energy is resolved, and the free-energy dependence remains undetermined. The absolute (111) values are about one third below those of Shuang et al. and the pure-Ni value is 24 % below DFT, but the composition trend agrees with the literature. The trends are therefore regarded as credible, and the absolute values as specific to the potential.

8. Data and Code Availability

All the input files, plotting scripts and numerical data for all 81 production runs (including `summary.csv`) are attached with the report on Google Classroom.

References

[1] Tran et al., Surface energies of elemental crystals, *Sci. Data* **3**, 160080 (2016).

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